

FACULTY MANUAL

Experiment 11

Power BI Dashboard (Basic): Data Import, Transformation and Basic Visualizations

Item	Details
Experiment No.	11
Title	Create dashboards using Power BI: data import, transformation, and basic visualizations
Tool	Microsoft Power BI Desktop (free)
Dataset	Sales_Data.csv (provided, 240 rows)
Duration	2 lab hours (approx. 100-120 minutes)
Level	Basic / Beginner

Note for faculty: Figures in this manual marked "illustrative" are reference renderings of the expected output, produced from the supplied dataset. Power BI screen layouts vary slightly by version (monthly release cycle). Boxes marked [Insert actual screenshot] are for you to paste real captures from your lab machine before circulating to students.

1. Aim

To create an interactive dashboard in Power BI by importing a dataset, cleaning and transforming it in Power Query, and building basic visualizations (cards, bar chart, line chart, donut chart and slicer).

2. Learning Outcomes

- Import data from a CSV file into Power BI Desktop.
- Clean and transform data using Power Query Editor (remove blanks, change data types, add a calculated column).
- Build basic charts and KPI cards and map fields to them.
- Add slicers/filters to make a dashboard interactive.
- Identify and handle common errors (invalid source, missing columns).

3. What is Power BI?

Microsoft Power BI is a Business Intelligence (BI) and data visualization tool. It connects to data from many places (Excel, CSV, databases, websites, cloud services), cleans and combines it, and turns it into interactive charts and dashboards that anyone can explore and share.

Power BI is a family of three parts:

Component	What it is	Used for
Power BI Desktop	Free Windows application	Importing, cleaning, modelling and building reports (used in this experiment)
Power BI Service	Online (cloud) portal at app.powerbi.com	Publishing, sharing and scheduling refresh of reports (needs a work/school account)
Power BI Mobile	App for Android / iOS	Viewing dashboards on phone or tablet

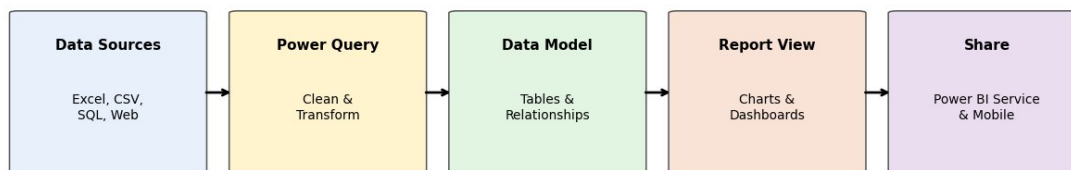


Figure A: Typical Power BI workflow (illustrative).

4. Why Use Power BI?

- Free to start: Power BI Desktop costs nothing to download and use for building reports.
- Handles far more data than a normal Excel sheet and stays fast.
- Connects to hundreds of data sources, so data from different systems can be combined in one report.
- No coding needed for basic work: drag and drop fields to build charts (advanced users can use DAX formulas).
- Interactive: click a chart or slicer and every visual updates, unlike static charts.

- Refreshable: when the source data changes, one click of Refresh updates the whole report because the cleaning steps are saved.
- Easy sharing through the Power BI Service and mobile app.
- High industry demand: BI and data analytics skills are widely listed in job requirements.

5. Applications of Power BI

Domain	Example dashboard
Sales & Retail	Sales by region/product, targets vs actual, top customers
Finance	Revenue, expense and profit tracking, budget variance
Education	Student attendance, result analysis, admissions trends
Healthcare	Patient footfall, bed occupancy, department-wise cost
Manufacturing	Production output, defect rate, machine downtime
HR	Headcount, attrition rate, recruitment pipeline
Marketing	Website traffic, campaign performance, social media reach
Government / Banking	Scheme monitoring, loan and transaction analysis

6. Power BI vs Excel

Feature	Excel	Power BI
Main purpose	Calculations, data entry, small analysis	Visual analytics and dashboards
Data size	Limited (about 10 lakh rows per sheet)	Millions of rows in the data model
Interactivity	Basic (filters, slicers, pivots)	Rich, cross-filtering across all visuals
Data refresh	Manual work often needed	Saved steps, one-click refresh
Sharing	File sharing by email	Published reports, mobile access
Cost	Paid desktop app (free web version)	Desktop free

Note: Excel is not required for this experiment because the dataset is a CSV file. Excel is useful for opening or editing the CSV, and Power BI can also import .xlsx files.

7. Requirements

- PC with Windows 10/11 (Power BI Desktop is Windows-only), 4 GB RAM or more.
- Power BI Desktop (download from Microsoft Store or powerbi.microsoft.com).
- Sales_Data.csv, supplied alongside this manual. Copy it to a folder such as C:\PowerBI_Lab\ on every machine before the session.

8. Where to Get Excel for Free

Desktop Microsoft Excel is a paid product, but there are legitimate free options. Avoid cracked software or unknown download sites, which often carry malware.

Option	How to get it	Notes
Excel for the web (best free option)	Go to office.com , sign in or create a free Microsoft account (any email works), and click Excel.	Runs in the browser, no installation, opens/saves .xlsx and .csv. Needs internet; fewer advanced features than desktop.
College / student Microsoft 365 Education	Sign in at office.com with your college email ID (if your institute has a licence).	May include the desktop apps depending on the institution. Ask the lab / IT department.
Excel mobile app	Install Microsoft Excel from Play Store / App Store.	Free with limits on small-screen editing.
Microsoft 365 free trial	Only through the official Microsoft 365 trial page on microsoft.com .	About 1 month, full desktop Excel; a payment method is usually requested. Cancel before it ends.
Free alternatives	LibreOffice Calc or Google Sheets.	Can open and edit CSV files for this experiment.

Steps to use Excel for the web

1. Open a browser and go to office.com.
2. Click Sign in, or Sign up for free to create a Microsoft account.
3. From the app list (waffle icon), click Excel.
4. Click New blank workbook, or upload Sales_Data.csv to OneDrive and open it.

[Insert actual screenshot] office.com home page with the Excel icon highlighted.

Faculty tip: Availability and offers change, so verify on microsoft.com before the lab session. For this experiment students can simply skip Excel and open the CSV in Notepad if needed.

9. Installing Power BI Desktop

Minimum system requirements: Windows 10 or later (64-bit recommended; Windows 7/8.1 are not supported), 4 GB RAM or more (8 GB recommended), display resolution of at least 1440 x 900, internet connection for download. Power BI Desktop does not run natively on macOS or Linux; Mac users can use the Power BI Service in a browser or a Windows virtual machine.

Method 1: Microsoft Store (recommended)

1. Open the Microsoft Store from the Start menu.
2. Search for "Power BI Desktop" (publisher: Microsoft Corporation).
3. Click Get / Install. Download is a few hundred MB and takes a few minutes.
4. After installation, click Open, or search for "Power BI Desktop" in the Start menu.
5. Close the welcome screen. You may sign in with a Microsoft account, or skip sign-in; this experiment needs no sign-in.

[Insert actual screenshot] Microsoft Store page of Power BI Desktop with the Get/Install button.

Advantage: The Store version does not need administrator rights and updates automatically each month.

Method 2: Direct download (if the Store is blocked)

1. Search for "Download Power BI Desktop" and open the official Microsoft Download Center page (microsoft.com/download).
2. Click Download, select the file PBIDesktopSetup_x64.exe (64-bit) and click Next.
3. Run the downloaded .exe file. Administrator rights are needed for this method.
4. Click Next, accept the licence agreement, keep the default install location and click Install.
5. Click Finish (keep "Launch Power BI Desktop" ticked).

[Insert actual screenshot] Power BI Desktop setup wizard with the licence agreement page.

Installation troubleshooting

Problem	Fix
Installer fails or freezes	Update Windows, run the installer as administrator, temporarily pause third-party antivirus
"App not compatible with this device"	Check Settings > System > About; install the 64-bit version on 64-bit Windows 10/11
Microsoft Store is blocked	Use Method 2 or ask the lab administrator
Some buttons or dialogs not visible	Set display scaling to 100% and resolution to at least 1440 x 900
Sign-in prompt appears	Click Skip / close; sign-in is not required for local report building

Tip: Power BI Desktop does not auto-save your .pbix file, so use File > Save regularly.

10. Dataset Description

Sales_Data.csv contains sample retail orders. Three rows are deliberately dirty (two blank Region values and one blank Sales value) so students can practise cleaning.

Column	Type	Description
OrderID	Text	Unique order number
OrderDate	Date	Date of order (2025)
Region	Text	North / South / East / West
Category	Text	Electronics / Furniture / Stationery
Quantity	Whole number	Units sold
UnitPrice	Whole number	Price per unit (INR)
Sales	Whole number	Quantity x UnitPrice
Profit	Whole number	Profit on the order (INR)

11. Theory in Brief

Power BI Desktop has three main views: Report (canvas for visuals), Data (table view) and Model (relationships). **Power Query Editor** is the ETL layer: every cleaning action is recorded as an "Applied Step" and replayed on each data refresh. **Visuals** are charts bound to fields via the Visualizations and Data panes. **Slicers** are on-canvas filters that cross-filter all other visuals.

12. Procedure (Step by Step)

Part A: Import the Dataset

1. **Open Power BI Desktop.** Close the welcome screen. You land on a blank Report view.
2. **Click Home > Get data > Text/CSV.**

[Insert actual screenshot] Home ribbon with Get data drop-down open and Text/CSV highlighted.

3. **Browse** to Sales_Data.csv and click Open.
4. **Review the preview window.** Check that column headers appear correctly. Note the blank (null) cells in Region and Sales.

Navigator - Sales_Data.csv (Preview)

OrderID	OrderDate	Region	Category	Quantity	UnitPrice	Sales	Profit
ORD-1229	02-01-2025	North	Stationery	4	2134	8536	1984
ORD-1075	02-01-2025	South	Stationery	7	3218	22526	4328
ORD-1013	02-01-2025	North	Electronic	4	1658	6632	448
ORD-1140	03-01-2025	West	Electronic	4	2850	11400	2055
ORD-1065	05-01-2025	East	Furniture	1	605	605	50
ORD-1142	06-01-2025	West	Furniture	3	2724	8172	1844
ORD-1193	10-01-2025	West	Furniture	6	4777	28662	4675
ORD-1197	11-01-2025	North	Furniture	7	3306	23142	6814
ORD-1049	13-01-2025	South	Stationery	7	3571	24997	3791
ORD-1181	14-01-2025	South	Stationery	8	1735	13880	2719
ORD-1047	16-01-2025	West	Furniture	5	1488	7440	1124
ORD-1090	16-01-2025	West	Furniture	8	2547	20376	5836

Display Options
[x] Sales_Data.csv

Load Transform Data Cancel

Figure 1: Navigator / preview window (illustrative). Choose "Transform Data", not "Load".

5. **Click Transform Data** (not Load) so that Power Query Editor opens and cleaning can be done before loading.

Expected result: Power Query Editor opens with the data shown. Applied Steps pane lists Source and Promoted Headers.

Part B: Transform (Clean) the Data

1. **Check data types.** OrderDate must show a calendar icon; Quantity, UnitPrice, Sales and Profit must show 123 (whole number). If not: click the icon in the column header and select the correct type.
2. **Remove blank rows in Region.** Click the drop-down arrow on the Region header, untick "(null)" / "(blank)" and click OK.
3. **Remove blank Sales rows.** Repeat the same on the Sales column (or use Home > Remove Rows > Remove Blank Rows).
4. **Add a calculated column.** Go to Add Column > Custom Column. Name it ProfitMargin%, enter the formula $[Profit] / [Sales] * 100$ and click OK. Set its type to Decimal number.

[Insert actual screenshot] Custom Column dialog with the ProfitMargin% formula entered.

5. **Verify Applied Steps.** The right-hand pane should now list all steps performed. Rows should reduce from 240 to 237.

Power Query Editor - Sales_Data								
APPLIED STEPS								
Source								
Promoted Headers								
Changed Type								
Removed Blank Rows								
Filtered Rows (no null Region)								
Added Custom: Profit Margin %								
OrderID	OrderDate	Region	Category	Quantity	UnitPrice	Sales	Profit	ProfitMargin%
ORD-1229	02-01-2025	North	Stationery	4	2134	8536	1984	23
ORD-1075	02-01-2025	South	Stationery	7	3218	22526	4328	19
ORD-1013	02-01-2025	North	Electronic	4	1658	6632	448	7
ORD-1140	03-01-2025	West	Electronic	4	2850	11400	2055	18
ORD-1065	05-01-2025	East	Furniture	1	605	605	50	8
ORD-1142	06-01-2025	West	Furniture	3	2724	8172	1844	23
ORD-1193	10-01-2025	West	Furniture	6	4777	28662	4675	16
ORD-1197	11-01-2025	North	Furniture	7	3306	23142	6814	29
ORD-1049	13-01-2025	South	Stationery	7	3571	24997	3791	15
ORD-1181	14-01-2025	South	Stationery	8	1735	13880	2719	20
ORD-1047	16-01-2025	West	Furniture	5	1488	7440	1124	15

Figure 2: Power Query Editor after cleaning, with Applied Steps (illustrative).

6. **Click Home > Close & Apply.** Data is now loaded into the model. Click the Data view icon (table icon at left) to confirm.

Faculty tip: Explain that transformations are non-destructive: the original CSV is never modified. Show students how deleting a step in Applied Steps reverses that action.

Part C: Create Basic Visualizations

Return to the Report view (top icon on the left bar). Ensure the Data pane (right) lists the table Sales_Data.

1. **Card: Total Sales.** Click a blank area of the canvas, choose the Card visual, drag Sales into Fields. Repeat for Profit, and for OrderID (set aggregation to Count) to show number of orders.
2. **Bar chart: Sales by Region.** Select Clustered column chart. Drag Region to X-axis and Sales to Y-axis.
3. **Line chart: Monthly trend.** Select Line chart. Drag OrderDate to X-axis (expand the date hierarchy and keep Month only) and Sales to Y-axis.
4. **Donut chart: Sales by Category.** Select Donut chart. Drag Category to Legend and Sales to Values.
5. **Slicer: Region.** Select the Slicer visual and drag Region into Field. Use Format pane > Slicer settings > Style to choose List.
6. **Format:** Add titles (Format > General > Title), align visuals neatly, and give the report a heading using Insert > Text box: "Sales Dashboard".

[Insert actual screenshot] Visualizations pane showing chart icons and the Fields well with Region and Sales assigned.

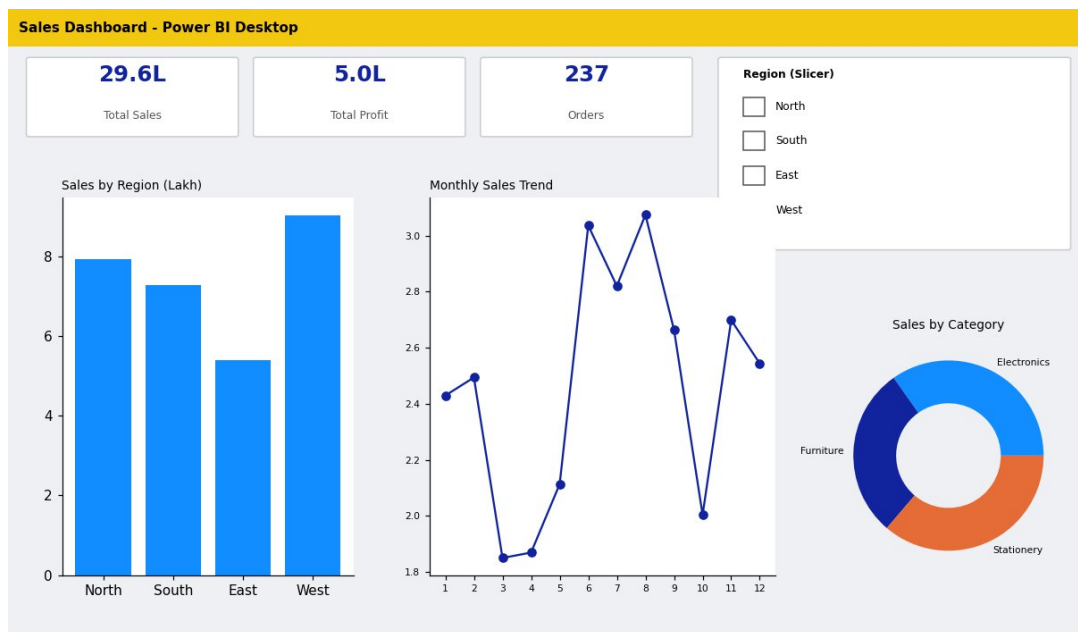


Figure 3: Completed dashboard, all regions (illustrative expected output).

Expected result: Three KPI cards (Total Sales approx. 29.6 lakh, Total Profit approx. 5.0 lakh, Orders = 237), a region-wise bar chart, a monthly trend line, a category donut and a Region slicer. Values are for the supplied dataset.

Part D: Apply Filters and Test Interactivity

1. **Click "South" in the Region slicer.** All visuals update to show only South-region data (cross-filtering).

2. **Use Ctrl + click** to select multiple regions. Click the same item again (or the eraser icon) to clear the filter.
3. **Click a slice of the donut chart.** Note how the bar chart and cards highlight/filter accordingly.
4. **Add a page-level filter (optional).** Drag Category into the Filters pane > "Filters on this page" and select Electronics.

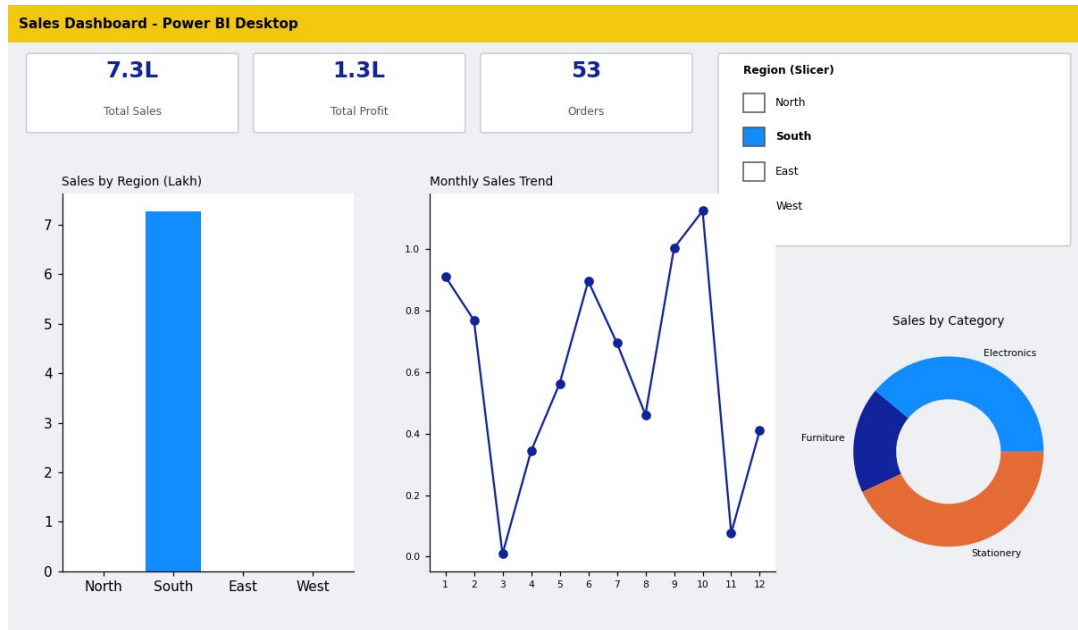


Figure 4: Dashboard after selecting Region = South (illustrative).

Expected result: Cards and all charts recalculate for the South region only, confirming interactivity.

Part E: Save the Work

1. **File > Save as** and name the file Exp11_<RollNo>.pbix.
2. **(Optional)** File > Export > PDF to submit a printable copy. Publishing to Power BI Service needs an institutional account and is outside basic scope.

13. Test Cases

Use these to verify the experiment. Students should record actual results in the last column.

Type	Input	Expected Output	Explanation	Actual
Success	Import dataset	Data loaded	Validates import	
Success	Create visuals	Charts displayed	Checks dashboard	
Success	Apply filters	Updated visuals	Tests interactivity	
Failure/Edge	Invalid data source	Load error	Detects issue	
Failure/Edge	Missing columns	Visualization error	Checks data mapping	

How to demonstrate the failure / edge cases

Test 4: Invalid data source. Rename the CSV file (e.g., Sales_Dta.csv) or move it, then click Home > Refresh (or Transform data > Source step). Power BI reports a DataSource.Error. Fix: Transform data > Data source settings > Change Source and point to the right file.

Power BI Desktop



Figure 5: Load error when the data source is invalid (illustrative).

Test 5: Missing columns. In the CSV copy, rename the Profit column header to Profits and refresh. The Profit card breaks, and Power Query shows "The column 'Profit' of the table wasn't found". Fix: rename the column back in Power Query or re-map the visual to the new field.

Report view - visual with missing field

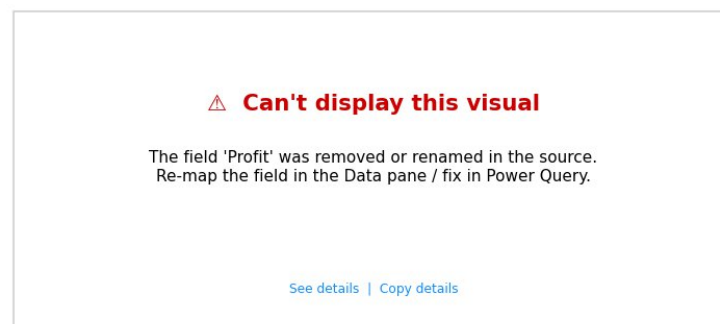


Figure 6: Visualization error when a mapped column is missing (illustrative).

Faculty tip: Always restore the original file after demonstrating the failure cases so the next batch is unaffected.

14. Common Student Mistakes and Fixes

Problem	Cause	Fix
Sales appears as text and cannot be summed	Wrong data type	Set to Whole number in Power Query, then Close & Apply
Chart shows "Count of Sales"	Wrong aggregation	Click the field in the Fields well and choose Sum
Dates appear as Year/Quarter/Month/Day	Auto date hierarchy	Expand hierarchy or click the arrow next to OrderDate and select "OrderDate" for raw dates
Slicer does not filter a visual	Interactions turned off	Format > Edit interactions and enable the filter icon
Changes not visible after transform	Forgot Close & Apply	Home > Close & Apply

15. Viva Questions

1. What is the difference between Load and Transform Data?
2. What are Applied Steps and why are they useful?
3. Difference between a measure and a calculated column?
4. What does a slicer do compared with a visual-level filter?
5. Which view would you use to create relationships between tables?
6. How do you refresh a report when the source file changes?

16. Suggested Evaluation Rubric (15 marks)

Criterion	Marks
Correct data import and preview	3
Data cleaning and custom column in Power Query	3
At least 4 correct visuals with titles	3
Working slicer / filter interactivity	3
Failure cases documented and saved file (.pbix)	3

17. Conclusion

Students imported a CSV dataset into Power BI, cleaned it with Power Query, built a multi-visual dashboard with cards, bar, line and donut charts, and verified interactivity using slicers. They also observed how invalid sources and missing columns affect reports.